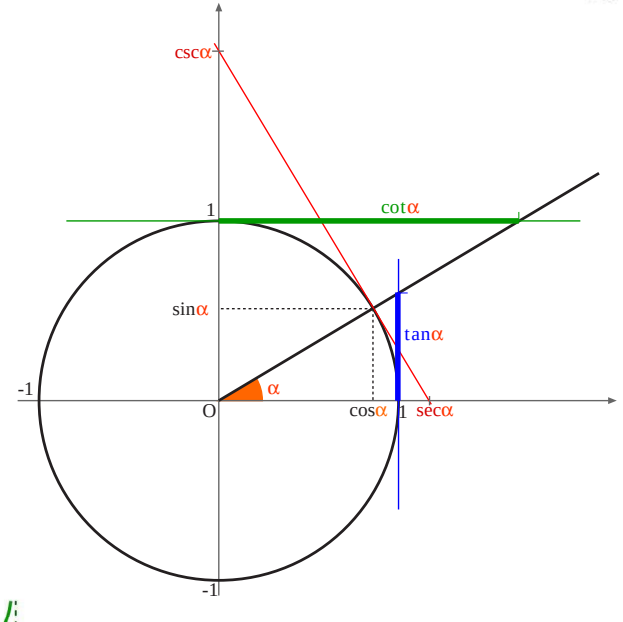
Aula 2: Funções trigonométricas

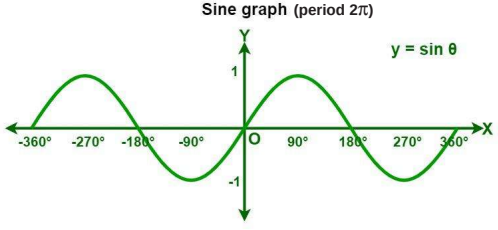
As funções trigonométricas seno, cosseno e tangente são definidas *geometricamente* no círculo trigonométrico, como estudado no Ensino Secundário.

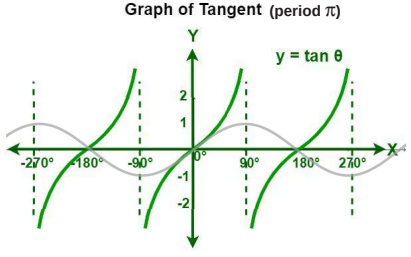
$$\operatorname{tg}\alpha = \frac{\operatorname{sen}\alpha}{\cos\alpha} \qquad \operatorname{cotg}\alpha = \frac{\cos\alpha}{\operatorname{sen}\alpha}$$

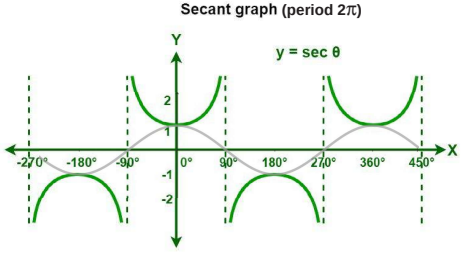
Outras funções trigonométricas: secante (sec) e cossecante (csc),

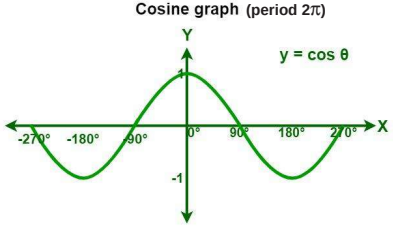
$$\sec\alpha = \frac{1}{\cos\alpha} \qquad \operatorname{csc}\alpha = \frac{1}{\operatorname{sen}\alpha}$$

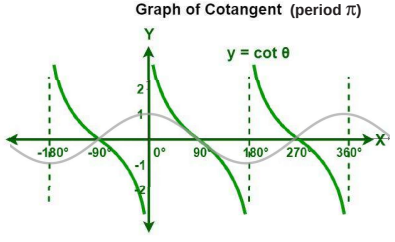


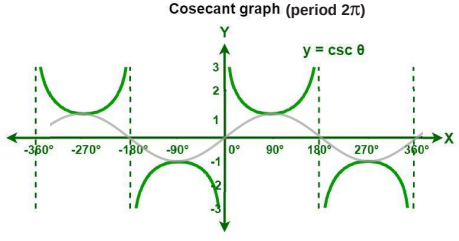












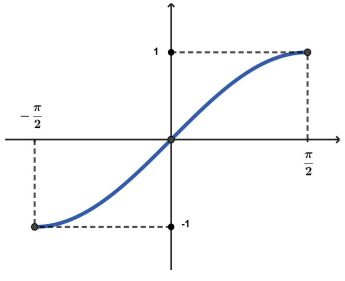
$$f(\alpha) = g(\tfrac{\pi}{2} - \alpha)$$
$$g(\alpha) = f(\tfrac{\pi}{2} - \alpha)$$

para $(f, g) = (\operatorname{sen}, \cos)$, $(\operatorname{tg}, \operatorname{cotg})$, $(\sec, \operatorname{csc})$

Aula 2: Funções trigonométricas inversas

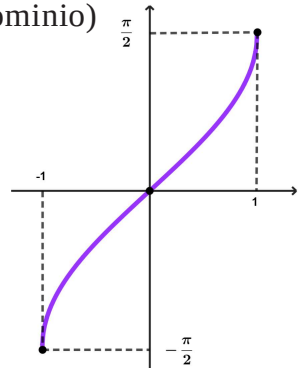
pf: parte de f ao subdominio (f restrita ao subdominio)

psen(x)
S(x)

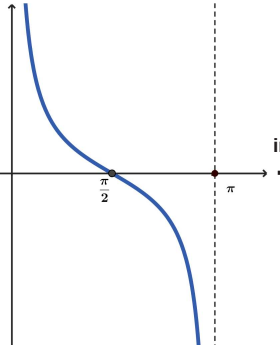


inversa

arcsen(x)

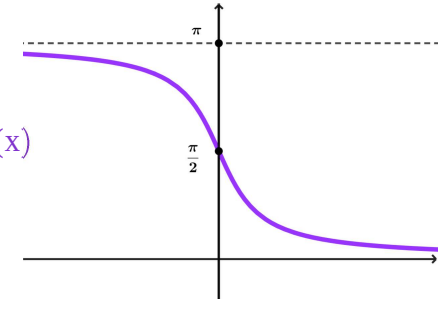


pcotg(x)

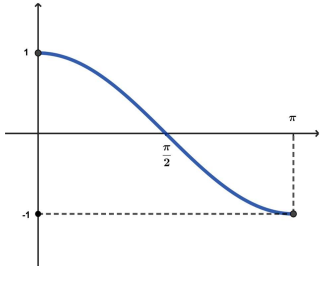


inversa

arccotg(x)

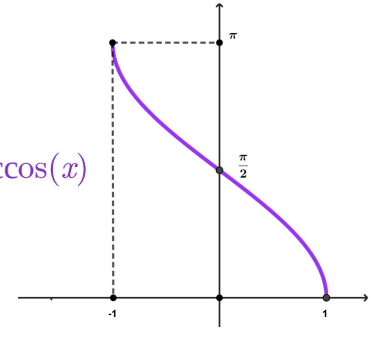


pcos(x)
C(x)

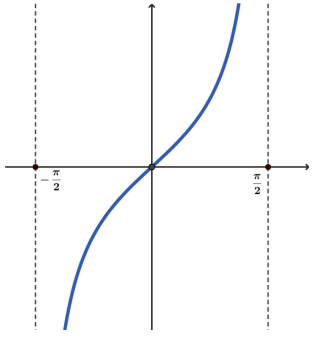


inversa

arccos(x)

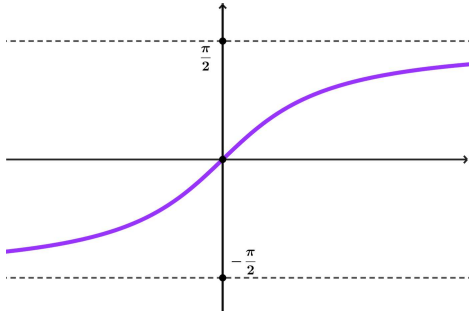


ptg(x)



inversa

arcrtg(x)



- Sejam f uma das funções trigonométricas analizadas e $arcf$ a inversa de pf.

Compondo pf com $arcf$ temos:

$$pf(arcf(x)) = x\,,\ \forall\, x \in D(arcf),$$

e $$arcf(pf(x)) = x\,,\ \forall\, x \in D(pf).$$

No entanto, compondo f com $arcf$, temos:

$$f(arcf(x)) = x\,,\quad \forall\, x \in D(arcf),$$

mas $$arcf(f(x)) \neq x\,,\ \forall\, x \in D(f).$$

Aula 3: Exercícios 1

Exercício 3.6

pg. 44

2. Considere a função dada por $f(x) = \arccos(e^x)$.
   (a) Determine o domínio e o contradomínio de f.
   (b) Determine, caso existam, os zeros de f.
3. Mostre que se $-1 \leq x \leq 1$ então $\operatorname{sen}(\arccos x) = \sqrt{1-x^2}$.

Exercício 3.7 Caracterize a função inversa da função f e estude-a quanto à continuidade, sendo $f(x) = \pi - \arccos(2x+1)$

Exercício 3.8 Mostre que as retas $y = x + \frac{\pi}{2}$ e $y = x - \frac{\pi}{2}$ são assíntotas ao gráfico da função $f(x) = x + \arctan x$.

Aula 3: Funções hiperbólicas

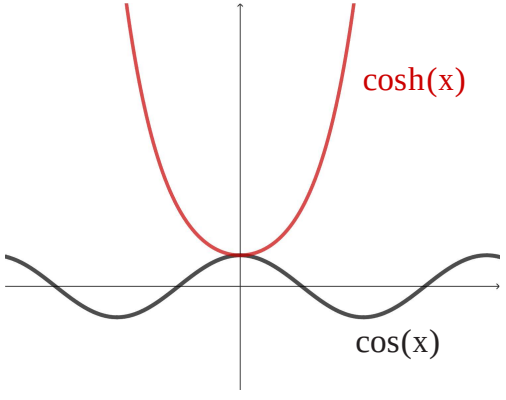
Fórmula de Euler: $\boxed{e^{ix} = \cos(x) + i\,\mathrm{sen}\,(x)}$ $\qquad (i^2 = -1)$

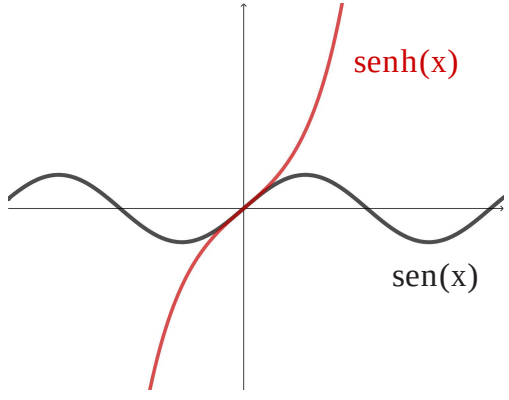
Daqui deduzimos:

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2} \quad , \quad \mathrm{sen}\,(x) = \frac{e^{ix} - e^{-ix}}{2i} \tag{1}$$

Eliminando o i nas formulas anteriores obtemos o coseno e o seno hiperbólicos:

$$\cosh(x) = \frac{e^{x} + e^{-x}}{2} \quad , \quad \mathrm{senh}\,(x) = \frac{e^{x} - e^{-x}}{2} \tag{2}$$





Aula 3: Fórmulas hiperbólicas e trigonométricas

$$\cosh(x) = \cos(ix)$$

$$\operatorname{senh}(x) = -i \operatorname{sen}(ix)$$

$$e^{ix} = \cos(x) + i \operatorname{sen}(x)$$

$$\cosh^2(x) - \operatorname{senh}^2(x) = 1$$

$$\operatorname{tgh}(x) := \frac{\operatorname{senh}(x)}{\cosh(x)} = \frac{e^{2x}-1}{e^{2x}+1}$$

$$\operatorname{cotgh}(x) := \frac{\cosh(x)}{\operatorname{senh}(x)} = \frac{e^{2x}+1}{e^{2x}-1}$$

$$\cosh(x+y) = \cosh(x)\cosh(y) + \operatorname{senh}(x)\operatorname{sen}(y)$$

$$\cosh(x-y) = \cosh(x)\cosh(y) - \operatorname{senh}(x)\operatorname{sen}(y)$$

$$\operatorname{senh}(x+y) = \operatorname{senh}(x)\cosh(y) + \cosh(x)\operatorname{senh}(y)$$

$$\operatorname{senh}(x-y) = \operatorname{senh}(x)\cosh(y) - \cosh(x)\operatorname{senh}(y)$$

$$\cosh(2x) = \cosh^2(x) + \operatorname{senh}^2(x)$$

$$\operatorname{senh}(2x) = 2\operatorname{senh}(x)\cosh(x)$$

$$(\cosh(x))' = \operatorname{senh}(x)$$

$$(\operatorname{senh}(x))' = \cosh(x)$$

$$\cos(x) = \cosh(ix)$$

$$\operatorname{sen}(x) = -i \operatorname{senh}(ix)$$

$$e^{x} = e^{i(-ix)} = \cosh(x) + i \operatorname{senh}(x)$$

$$\cos^2(x) + \operatorname{sen}^2(x) = 1$$

$$\operatorname{tg} x = \frac{\operatorname{sen}(x)}{\cos(x)}$$

$$\operatorname{cotg} x = \frac{\cos(x)}{\operatorname{sen}(x)}$$

$$\cos(x+y) = \cos(x)\cos(y) - \operatorname{sen}(x)\operatorname{sen}(y)$$

$$\cos(x-y) = \cos(x)\cos(y) + \operatorname{sen}(x)\operatorname{sen}(y)$$

$$\operatorname{sen}(x+y) = \operatorname{sen}(x)\cos(y) + \cos(x)\operatorname{sen}(y)$$

$$\operatorname{sen}(x-y) = \operatorname{sen}(x)\cos(y) - \cos(x)\operatorname{sen}(y)$$

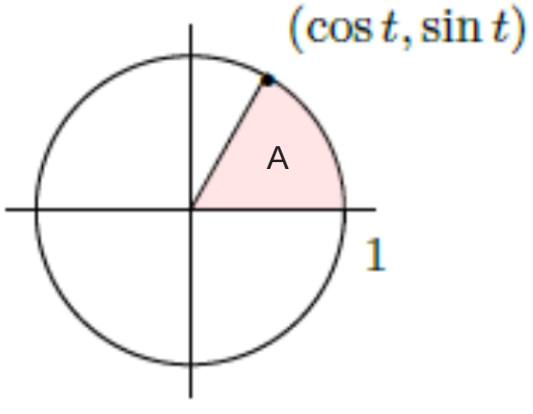
$$\cos(2x) = \cos^2(x) - \operatorname{sen}^2(x)$$

$$\operatorname{sen}(2x) = 2\operatorname{sen}(x)\cos(x)$$

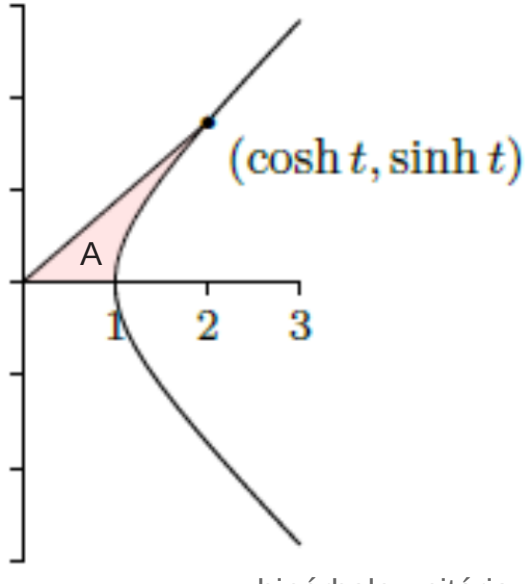
$$(\cos(x))' = -\operatorname{sen}(x)$$

$$(\operatorname{sen}(x))' = \cos(x)$$

Aula 3: Semelhanças



circunferência unitária

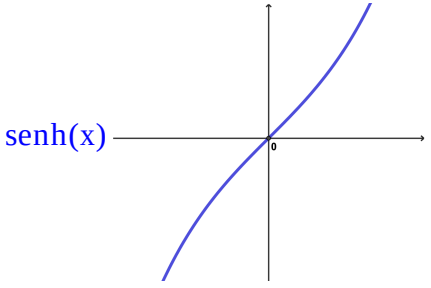


hipérbole unitária

Área $A = \frac{t}{2}$

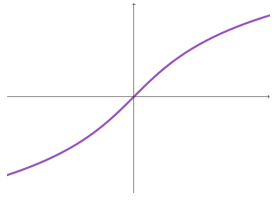
Aula 3: Funções hiperbólicas inversas

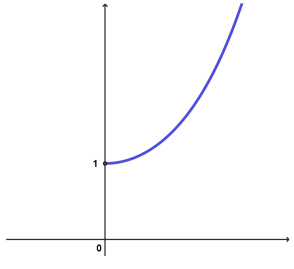
pf: parte de f ao subdominio (f restrita ao subdominio)



inversa

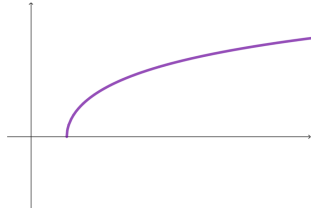
$\overset{-1}{\text{senh}}(x) = \ln(x+\sqrt{x^2+1})$

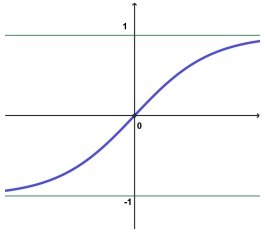




inversa

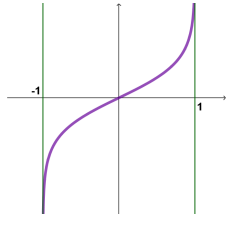
$\overset{-1}{\text{pcosh}}(x) = \ln(x+\sqrt{x^2-1})$

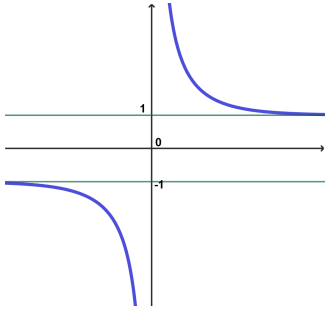




inversa

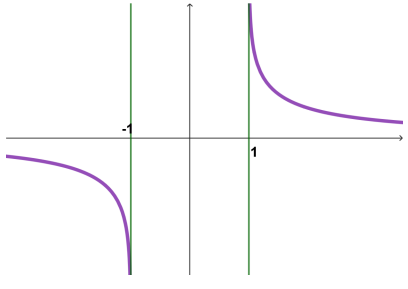
$\overset{-1}{\text{tgh}}(x) = \frac{1}{2}\ln(\frac{1+x}{1-x})$





inversa

$\overset{-1}{\text{cotgh}}(x) = \frac{1}{2}\ln(\frac{x+1}{x-1})$



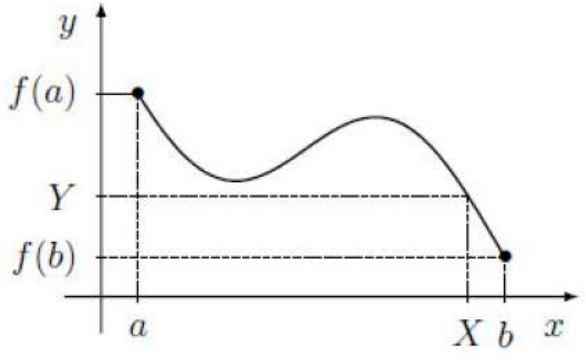
Nota: (1) À semelhança das funções trigonométricas "inversas", também aqui é habitual designar-se por arcf à inversa de f (f=senh, tgh, cotgh) ou pf (f=cosh).

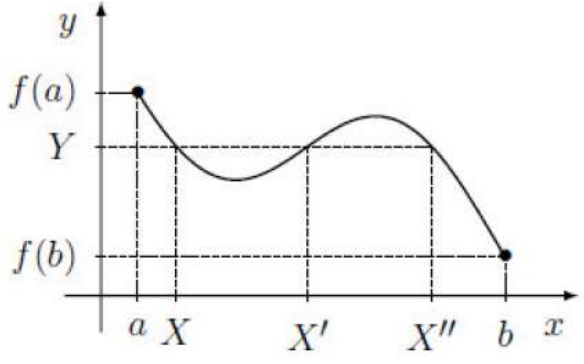
Neste último caso, $\text{arccosh} = \overset{-1}{\text{pcosh}}(x)$.

(2) $\cosh(\overset{-1}{\text{pcosh}}(x)) = x$ mas $\overset{-1}{\text{pcosh}}(\cosh(x)) = x$ $(=|x|)$.

Aula 3: Teoremas sobre funções contínuas - Teorema de Bolzano

Teorema 4.1. ***(Teorema de Bolzano ou dos valores intermédios)*** *Se f é uma função contínua num intervalo $[a,b], a < b$, e $f(a) < Y < f(b)$ ou $f(b) < Y < f(a)$ então existe $X \in]a,b[$ tal que $f(X) = Y$.*





Corolário 1. *Se f é contínua em $[a,b]$ e $f(a) \cdot f(b) < 0$ então existe $x_0 \in]a,b[$ tal que $f(x_0) = 0$.*

Exemplo 4.1. A equação $\operatorname{sen} x + 2x - 1 = 0$ tem pelo menos uma solução em $\mathbb{R}$.

(estritamente monótona)

Corolário 2. *Seja I um intervalo qualquer de $\mathbb{R}$ e $f : I \to \mathbb{R}$ uma função contínua. Então $f(I)$ é um intervalo.*

Um ponto é um intervalo degenerado

Aula 3: Extremos locais e globais

Seja $f : D_f \to \mathbb{R}$ com contradomínio CD_f. Um ponto $c \in D_f$ é

- ponto de *máximo* (*mínimo*) *global* se $f(c)$ é o máximo (mínimo) de CD_f;
- ponto de *máximo* (*mínimo*) *local* se existe uma vizinhança $\mathcal{V}(c)$ tal que c é ponto de máximo (mínimo) global da restrição da função f ao conjunto $D \cap \mathcal{V}(c)$.

Um ponto c de máximo ou de mínimo local (global) diz-se *ponto de extremo local* (*global*) de f. Ao valor $f(c)$ chama-se *extremo local* (*global*) de f.

ou extremante

Um ponto de extremo c da função f diz-se *extremo estrito* quando $f(x) \neq f(c)$ para todo o x diferente de c numa vizinhança de c.

Exercício 4.1 Encontre extremos e pontos de extremo locais e globais da função definida em $[-4, 6[$ pelo gráfico representado na figura 4.2, indicando os extremos estritos:

